

Bureau Dataset

Comprehensive Multi-Perspective Analysis Report

[Business](#) | [Product](#) | [Data Quality](#) | [Data Science](#)

Real-World Challenges | Integration Strategy | Cross-Dataset Synergies

Home Credit Default Risk — bureau.csv

NUS — Institute for Financial Risk Management (IFRM)

Table of Contents

1. Executive Summary
2. Dataset Overview & Schema
3. Real-World Challenges Analysis
4. Credit Portfolio Composition
5. Credit Amounts & Financial Exposure
6. Credit Timeline & Age Analysis
7. Default Rate Deep-Dive
8. Credit Type Risk Profiling
9. Overdue, Delinquency & Prolongation
10. Debt-to-Credit Ratio Analysis
11. Missing Values & Data Quality
12. Integration with application_train.csv
13. Cross-Dataset Synergies
14. Feature Engineering Playbook
15. Business Recommendations
16. Conclusions

1. Executive Summary

The **bureau.csv** dataset is the **central nervous system of external credit data** in the Home Credit feature suite. It contains **1,716,428 credit records** from **305,811 applicants** across **15 credit types**, capturing every credit that Home Credit's applicants have ever held at **other financial institutions**. This is not just behavioral data — it is a complete **credit portfolio snapshot** including amounts, status, overdue history, and timeline.

Dimension	Key Finding	Impact
Scale	1,716,428 credits from 305,811 applicants	Mean 5.6 credits per applicant; max 116
Coverage	263,491 of 307,511 train apps (85.7%)	Strong coverage; 14.3% of train apps have NO bureau data
Credit Status	Closed: 62.9%, Active: 36.7%, Sold: 0.4%, Bad debt: 0.001%	Portfolio majority is historical; 1/3 currently active
Credit Types	Consumer: 72.9%, Card: 23.4%, Car: 1.6%, Mortgage: 1.1%	Dominated by consumer credit; revolving credit significant
Delinquency	Current overdue: 0.25% of credits; max overdue: 2,792 days	Low current delinquency but extreme outliers exist
Financial Scale	Total credit median: 125K; max: 585M; debt median: 0	Huge variance in exposure; most historic credits repaid
Default Signal	Without bureau: 10.12% default; With bureau: 7.73%	Having bureau history itself is a protective factor

Strategic Importance: Bureau.csv is the **most information-rich supplementary dataset**. With 17 columns spanning credit amounts, timeline, overdue status, credit type, and active/closed status, it provides a panoramic view of each applicant's external credit life. It is the **parent table for bureau_balance.csv** (monthly status) and the **only source** of credit amounts and types at other institutions. Features derived from bureau data consistently rank among the top predictors in champion models.

2. Dataset Overview & Schema

Each row represents **one external credit** that an applicant held at another financial institution, as reported by credit bureaus to Home Credit. The dataset has **17 columns**:

Column	Type	Null %	Description
SK_ID_CURR	int64	0%	Applicant ID — direct join to application_train
SK_ID_BUREAU	int64	0%	Unique bureau credit ID — joins to bureau_balance
CREDIT_ACTIVE	object	0%	Status: Closed (62.9%), Active (36.7%), Sold (0.4%), Bad debt
CREDIT_CURRENCY	object	0%	Currency (99.9% = currency 1)
DAYS_CREDIT	int64	0%	Days since credit opened (negative = past; min=-2922)
CREDIT_DAY_OVERDUE	int64	0%	Current days overdue (0 for 99.75% of credits)
DAYS_CREDIT_ENDDATE	float64	6.1%	Expected end date (days); positive = future, neg = past
DAYS_ENDDATE_FACT	float64	36.9%	Actual end date (days); NULL for active credits
AMT_CREDIT_MAX_OVERDUE	float64	65.5%	Max amount of overdue ever recorded
CNT_CREDIT_PROLONG	int64	0%	Times credit was prolonged (99.5% = 0)
AMT_CREDIT_SUM	float64	~0%	Total credit amount; mean=355K, max=585M
AMT_CREDIT_SUM_DEBT	float64	15.0%	Current outstanding debt; NULL mainly for closed credits
AMT_CREDIT_SUM_LIMIT	float64	34.5%	Credit limit (revolving); 0 for most non-revolving
AMT_CREDIT_SUM_OVERDUE	float64	0%	Current overdue amount; 0 for 99.76% of credits
CREDIT_TYPE	object	0%	Type: Consumer (73%), Card (23%), Car (1.6%), etc.
DAYS_CREDIT_UPDATE	int64	0%	Days since last info update from bureau
AMT_ANNUITY	float64	71.5%	Credit annuity amount; heavily missing

Join architecture: bureau.csv occupies a **pivotal position** in the data model. It joins **directly** to application_train via SK_ID_CURR (one-to-many: one applicant → many credits), and is the **parent** of bureau_balance via SK_ID_BUREAU (one-to-many: one credit → many monthly records). This makes it the **bridge** between applicant-level data and granular monthly external behavior.

3. Real-World Challenges Analysis

Challenge 1: Heavy Missing Values — Selective Reporting

Seven columns have significant null values, ranging from AMT_CREDIT_SUM (near zero) to AMT_ANNUITY (71.5% missing). The missing values are **not random** — they follow credit type and status. AMT_ANNUITY is missing mostly for closed credits (the annuity information wasn't retained after closure). DAYS_ENDDATE_FACT is naturally NULL for active credits (they haven't ended yet). AMT_CREDIT_MAX_OVERDUE is missing for 65.5% — credits that never had overdue simply don't have this field populated.

Data Science Principle: Treat missingness as a **feature, not just a gap**. Create binary flags: has_annuity_info, has_overdue_history, has_end_date. These flags carry information about credit type, reporting depth, and credit lifecycle stage.

Challenge 2: Extreme Outliers Across Financial Columns

AMT_CREDIT_SUM ranges from 0 to 585,000,000 (585M) — a ratio of infinity. AMT_CREDIT_MAX_OVERDUE goes up to 115,987,185 (116M). AMT_ANNUITY reaches 118,453,424. These extreme values likely represent corporate credits or reporting errors. The median credit amount is 125K while the max is 585M — a 4,680x gap. Such extremes distort mean-based aggregations and can dominate model training.

Treatment: Use **median/percentile aggregations** instead of mean. Apply winsorization at 99th percentile. Log-transform large monetary amounts. Create separate features for 'has_extreme_credit' (> 10M) as a credit type proxy.

Challenge 3: One-to-Many Aggregation Complexity

Each applicant has **mean=5.6, max=116** credits. Aggregating 17 columns × multiple aggregation functions × per applicant creates a feature explosion. With 5 aggregation functions (mean, max, min, sum, count) × 12 numeric columns = 60+ features before considering credit type segmentation or temporal splits.

Risk of over-engineering: Blindly generating all possible aggregations leads to multicollinearity and noise. Prioritize features with known business meaning: total_debt, max_overdue, n_active, credit_age_range. Use feature importance from initial models to prune.

Challenge 4: Temporal Ambiguity — DAYS_CREDIT

DAYS_CREDIT measures days between credit opening and application date (always negative or zero). Credits from 8 years ago may have very different relevance compared to credits opened last month. Historical defaults that were resolved 5 years ago should carry less weight than recent delinquency — but the dataset treats them equally unless you engineer temporal decay.

Challenge 5: Credit Status Imbalance

Bad debt — the strongest default signal — has only **21 records** (0.001%). **Sold** credits (6,527, 0.4%) are also rare. This extreme imbalance means these statuses are nearly invisible in aggregations but carry enormous predictive power when present. A single 'Bad debt' credit among 10 others changes the risk profile dramatically.

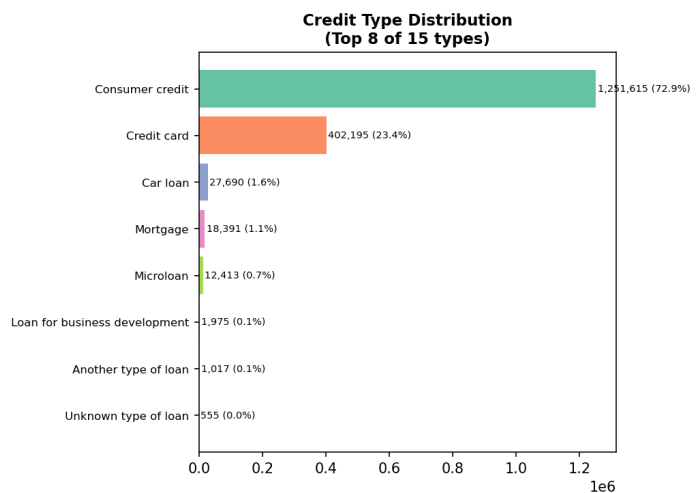
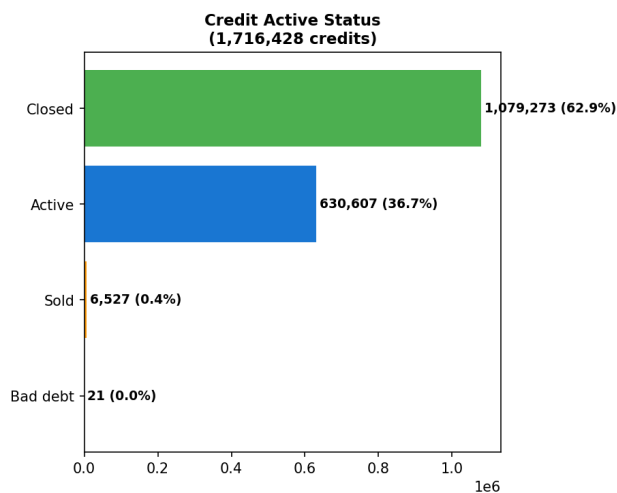
Solution: Create **binary flags** (has_bad_debt, has_sold_credit) rather than proportions. These are rare but powerful features.

Challenge 6: Cross-Currency Comparisons

99.9% of credits are in 'currency 1', but 1,408 credits are in other currencies. Summing credit amounts across currencies without conversion creates distorted exposure measures. Features should either normalize to a

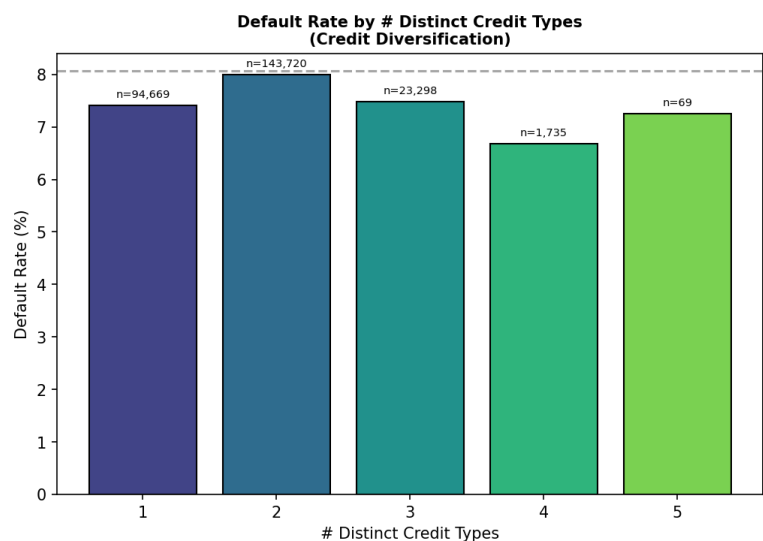
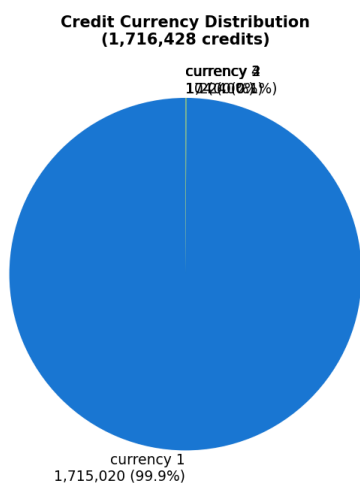
common currency or segment by currency.

4. Credit Portfolio Composition



Credit Active Status: The portfolio is predominantly historical — **62.9% Closed**, meaning the applicant has already fully repaid. **36.7% Active** represents ongoing obligations. **Sold credits (0.4%)** were transferred to collection agencies or other institutions — a yellow flag. **Bad debt (21 records)** is vanishingly rare but devastating: applicants with any bad debt show a **20% default rate** vs 7.7% baseline.

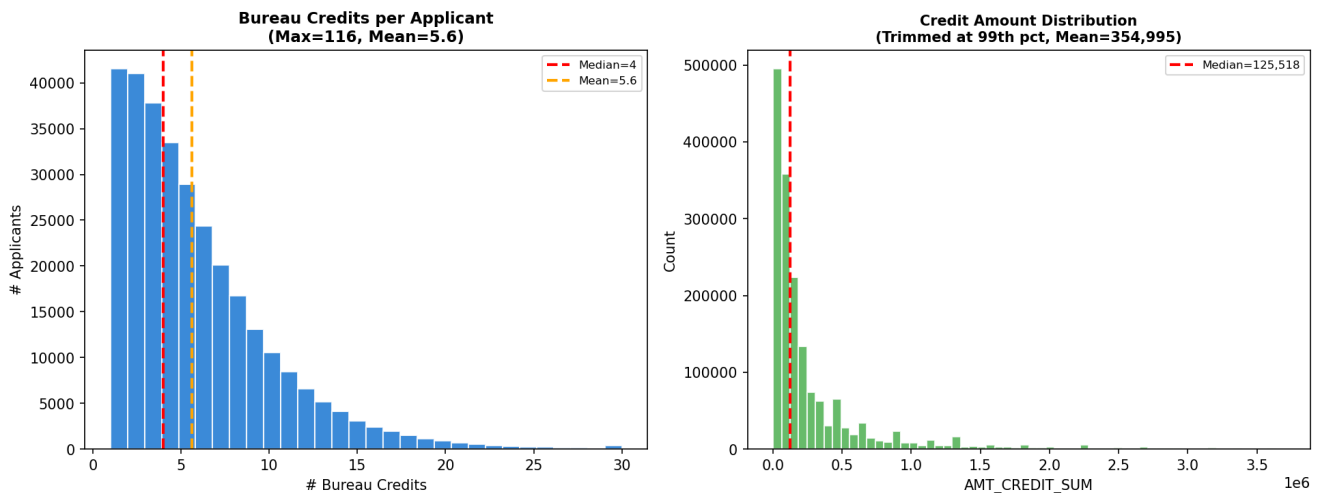
Credit Type Dominance: Consumer credit (72.9%) and Credit cards (23.4%) comprise 96.3% of all records. The remaining 3.7% covers 13 different types including Car loans, Mortgages, Microloans, and Business loans. This long-tail distribution means rare credit types are underrepresented in training — features derived from them will be noisy unless carefully handled.



Currency: Near-total dominance of 'currency 1' (99.9%). The few non-standard currency credits may indicate international borrowing or speciality products — worth flagging but not aggregating separately due to sample size.

Credit Diversification: Applicants with more distinct credit types show somewhat higher default rates, peaking at 5+ types. Over-diversification in external credit may signal financial stretching — borrowing from many product types simultaneously to meet obligations.

5. Credit Amounts & Financial Exposure



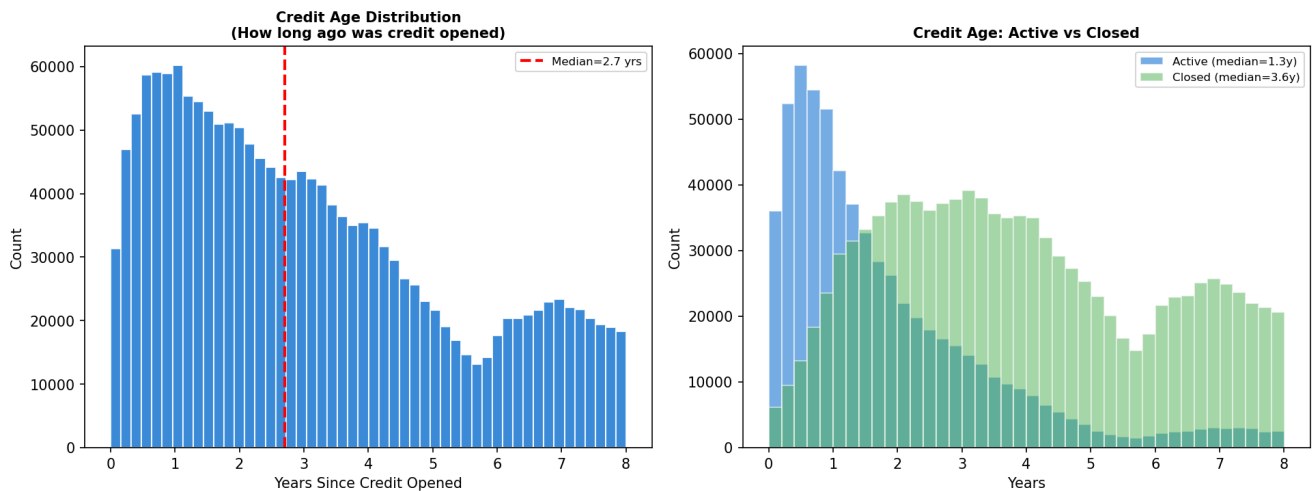
Credits per Applicant: Median 4, Mean 5.6, heavily right-skewed. Most applicants have 2-8 external credits, but a significant tail extends to 116. Applicants with 1 credit show the highest default rate (8.6%) — they may be new to the credit market. The '20+' group also shows elevated default (11.2%), suggesting over-leveraging.

Metric	AMT_CREDIT_SUM	AMT_CREDIT_SUM_DEBT	AMT_CREDIT_SUM_LIMIT
Mean	354,995	137,085	6,230
Median	125,519	0	0
Max	585,000,000	170,100,000	4,705,600
Null %	~0%	15.0%	34.5%
Zero %	3.9%	Most closed = 0	93.4% = 0
Meaning	Total credit size	Outstanding debt	Available limit (revolving only)

Key observation: AMT_CREDIT_SUM_DEBT has median = 0, meaning most credits have zero outstanding debt (they're closed and repaid). For active credits, the debt distribution is meaningful. AMT_CREDIT_SUM_LIMIT is only relevant for revolving credit (credit cards); 93.4% of records show zero because they're installment-type credits.

Business Insight: The total external credit exposure (sum of AMT_CREDIT_SUM across active credits) is a crucial metric for **debt service capacity**. An applicant requesting a 500K loan from Home Credit while carrying 2M in active external debt has a fundamentally different risk profile than one with zero external exposure. This information is **unavailable** from any internal Home Credit dataset.

6. Credit Timeline & Age Analysis



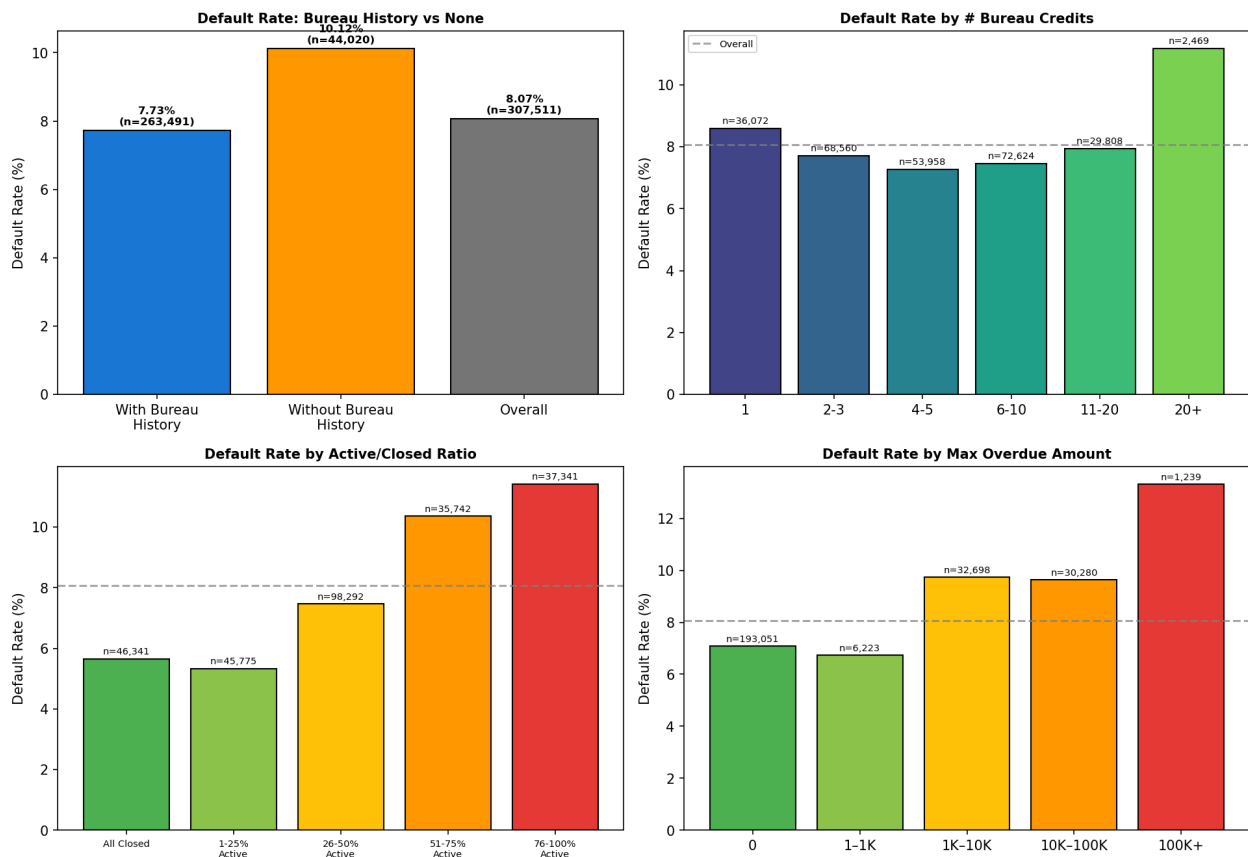
Credit age distribution: External credits span from **same-day (0 days)** to **8.0 years ago**. The median credit was opened 2.7 years before the application. The distribution shows a gradual build-up of credit history, with a slight concentration of credits opened 1-3 years before application.

Active vs Closed age: Active credits are younger (median ~1.3 years) while closed credits are older (median ~3.6 years). This is natural — older credits have had time to reach maturity. The overlap region (1-3 years) is where both active and closed credits coexist, representing the 'active lifecycle zone'.

Metric	DAYS_CREDIT (opened)	DAYS_CREDIT_ENDDATE (expected end)	DAYS_ENDDATE_FACT (actual end)	DAYS_CREDIT_UPDATE (last update)
Mean	-1,142 days (3.1 years ago)	511 days (1.4 yrs future)	-1,017 days (2.8 yrs ago)	-594 days (1.6 yrs ago)
Median	-987 days	-330 days	-897 days	-395 days
Range	-2922 to 0	-42060 to 31199	-42023 to 0	-41947 to 372
NULL %	0%	6.1%	36.9%	0%
Meaning	When credit was opened	When expected to end	When actually closed	Last bureau report date

Temporal feature engineering opportunity: The gap between expected end date and actual end date ($\text{DAYS_CREDIT_ENDDATE} - \text{DAYS_ENDDATE_FACT}$) reveals whether credits were **closed early** (positive = good) or **extended/defaulted** (negative = concerning). Credits that ended well before the expected date show the applicant prepaid — a strong positive signal.

7. Default Rate Deep-Dive

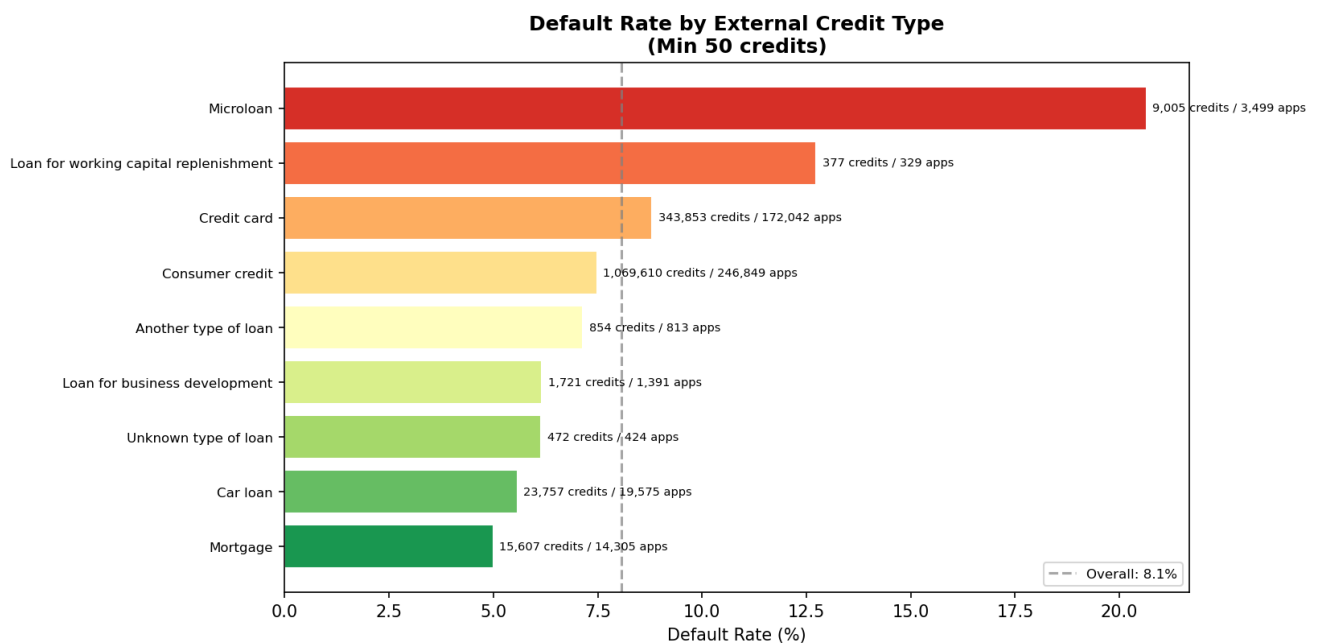


Key findings from multi-dimensional default analysis:

- * **Bureau presence:** Applicants WITH bureau history show **7.73% default vs 10.12%** without. Having external credit history itself is a **protective factor** — it indicates credit market participation and experience. The 14.3% of applicants with no bureau record are disproportionately risky.
- * **Number of credits (U-curve):** Single-credit applicants (8.6% default) and 20+ credit applicants (11.2%) both show elevated risk. The sweet spot is 4-10 credits (~7.3-7.5%), indicating **moderate credit experience** without over-leverage.
- * **Active/closed ratio (monotonic):** The strongest signal. All-closed portfolios show **5.7% default** — these are experienced borrowers who've completed obligations. As active ratio increases: 25% active → 5.3%, 50% → 7.5%, 75% → 10.4%, all-active → **11.4%**. A perfectly monotonic relationship revealing that **current external obligations** correlate with risk.
- * **Max overdue amount:** Zero overdue = 7.1% default. Any overdue 10K-100K = 9.6%. Overdue 100K+ = **13.3% default**. The magnitude of historical overdue directly predicts future default, with a clear dose-response relationship.

Actionable Insight: The active/closed ratio is the single most discriminating bureau-level feature. An applicant whose external portfolio is 100% active (no completed credits) defaults at **2× the rate** of one with all credits closed. This reflects both **maturity** (experience completing obligations) and **current load** (no active debts = no external stress).

8. Credit Type Risk Profiling



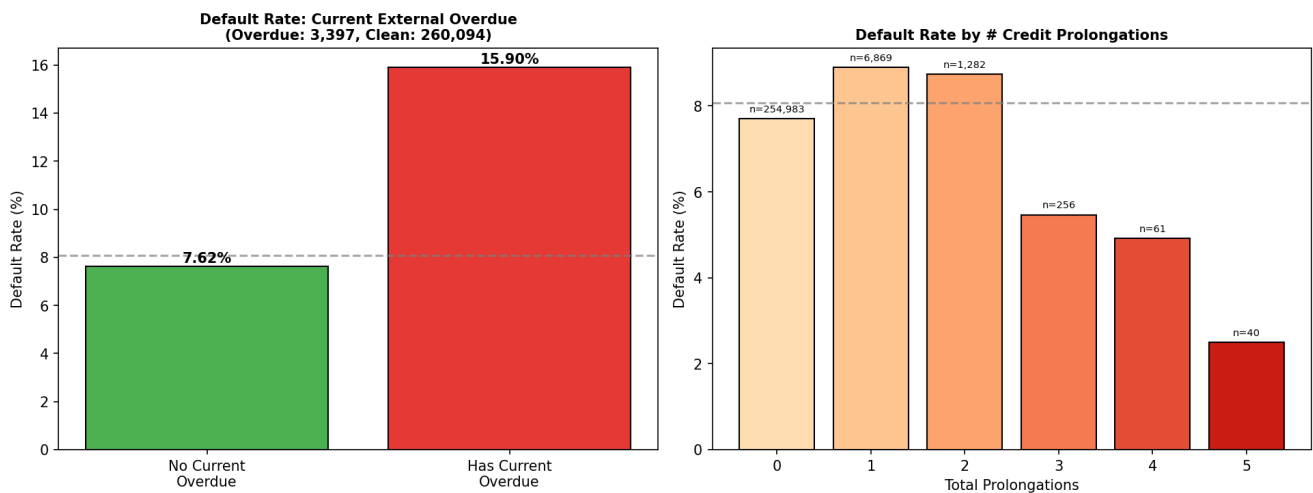
Default rates vary dramatically across credit types:

Credit Type	# Credits	Default Rate	Risk Level	Business Interpretation
Microloan	9,005	20.6%	VERY HIGH	Distressed borrowers; predatory lending indicator; 2.5× baseline
Working capital loan	377	12.7%	HIGH	Business cash flow issues; often last-resort borrowing
Credit card	343,853	8.8%	ELEVATED	Revolving credit users; potential debt spiraling
Consumer credit	1,069,610	7.5%	BASELINE	Standard installment credit; largest population
Another type	854	7.1%	MODERATE	Miscellaneous products; unclear risk pattern
Car loan	23,757	5.6%	LOW	Secured credit; collateral reduces loss-given-default
Mortgage	15,607	5.0%	LOWEST	Most vetted applicants; strong collateral; longest track record

Microloan as distress signal: Applicants with microloans default at 20.6% — nearly **3× the baseline**. Microloans are typically small, short-term, high-interest products used by borrowers who can't access traditional credit. Having a microloan in the bureau record is one of the strongest individual risk signals in the entire dataset.

Product strategy: Credit type composition reveals the applicant's position in the credit spectrum. A portfolio of mortgages + car loans signals established, asset-backed borrowing. A portfolio of microloans + consumer credit signals potential financial distress. Create features: `has_microloan`, `has_mortgage`, `pct_secured_credit`, `pct_revolving_credit`.

9. Overdue, Delinquency & Prolongation



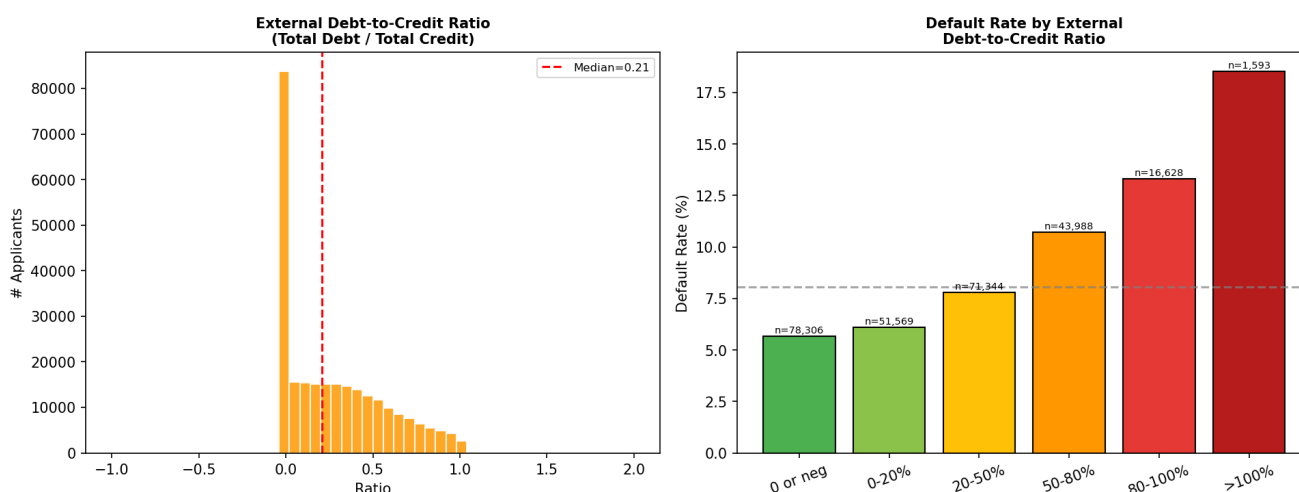
Current overdue status: Only **0.25% of credits** (4,217 out of 1,716,428) are currently past due. The maximum is 2792 days (~7.7 years) — clearly a write-off that was never formally closed. While current overdue is rare, it's an extremely strong signal: applicants with **any** current external overdue show significantly higher Home Credit default rates.

Credit prolongation: Only 9,114 credits (0.53%) were ever prolonged. Some were prolonged up to 9 times. Prolongation indicates the borrower couldn't complete the original term — they needed more time. Default rate increases with prolongation count, suggesting that needing extensions at external institutions predicts inability to repay Home Credit.

Metric	CREDIT_DAY_OVERDUE (Days)	CREDIT_MAX_OVERDUE (Amount)	CREDIT_SUM_OVERDUE (Current Amount)
Non-zero %	0.25%	20.5% (of non-null)	0.24%
Mean (when > 0)	333 days	18,669	15,650
Max	2,792 days	115,987,185	3,756,681
Null %	0%	65.5%	0%
Meaning	Days currently past due	Worst historical overdue amount	Amount currently past due

Three-dimensional overdue view: Bureau.csv provides three complementary overdue measures: (1) Current days overdue — snapshot, (2) Max historical overdue amount — worst-ever, (3) Current overdue amount — active balance. Used together, these capture the **severity, history, and current state** of delinquency at external institutions.

10. Debt-to-Credit Ratio Analysis



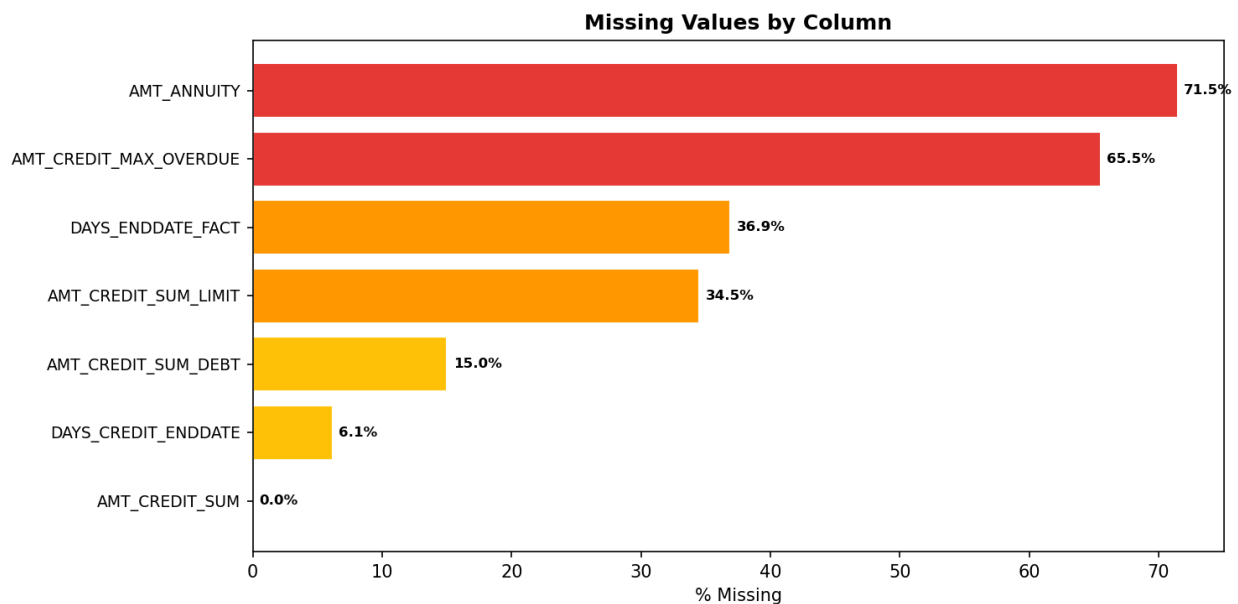
External Debt-to-Credit Ratio (Total Debt / Total Credit) is the applicant's external leverage measure. This captures how much of their total external credit capacity is currently being used as outstanding debt.

Distribution: Median ratio is 0.21 — most applicants have repaid most of their external credit. The distribution peaks at 0 (fully repaid) with a secondary mode near 0.3-0.5 (active partial debt).

Default rate relationship: Clear monotonic increase with leverage. Applicants with 0% debt-to-credit (fully repaid) show the lowest default. Those at 80-100% (nearly maxed out) show progressively higher default. Applicants over 100% (debt exceeds original credit — penalties, interest accumulation) are the highest risk.

Business Application: The external debt-to-credit ratio is analogous to **credit utilization** in traditional scoring. High utilization signals financial stress. When combined with Home Credit's own loan amount, it reveals the applicant's **total debt load** across all institutions — a metric that no single-institution analysis can provide. This is Home Credit's unique advantage from having bureau data.

11. Missing Values & Data Quality



Column	Null %	Root Cause	Treatment Strategy
AMT_ANNUIITY	71.5%	Not reported for closed credits or certain types	Flag has_annuity; impute with type-specific medians for active
AMT_CREDIT_MAX_OVERDUE	65.5%	Not tracked when credit never had overdue	NULL → 0 (no overdue history); create has_overdue_history flag
DAYS_ENDDATE_FACT	36.9%	Active credits haven't ended yet; naturally NULL	NULL → flag is_active; no imputation needed
AMT_CREDIT_SUM_LIMIT	34.5%	Only exists for revolving credit (cards, lines)	NULL → 0 for non-revolving; create is_revolving flag
AMT_CREDIT_SUM_DEBT	15.0%	Closed/paid credits may not report remaining debt	NULL → 0 for closed credits; flag for active NULL cases
DAYS_CREDIT_ENDDATE	6.1%	No defined end date for some credit types	Impute with median by type; flag has_end_date
AMT_CREDIT_SUM	~0%	13 records — data entry errors or edge cases	Impute with median by type; negligible impact

Key principle: In bureau.csv, missing values are **informative**. AMT_ANNUIITY missing ≈ closed credit. DAYS_ENDDATE_FACT missing ≈ active credit. AMT_CREDIT_MAX_OVERDUE missing ≈ clean payment history. Converting NULLs to flags captures this information explicitly.

Additional quality issues:

- * **Negative AMT_CREDIT_SUM_DEBT:** Some records show negative debt (min = -4,705,600). This likely represents overpayments or accounting adjustments. Treatment: clip at 0 or create a flag for negative debt.
- * **DAYS_CREDIT_ENDDATE extreme values:** Range includes -42,060 (115 years ago) and 31,199 (85 years in future). These are clearly data errors or placeholder values. Clip at reasonable bounds (±10 years) or flag.
- * **AMT_CREDIT_SUM extreme outliers:** Max 585M vs median 125K — a 4,680× ratio. These extreme values distort mean-based aggregations. Use median or winsorized mean for per-applicant features.

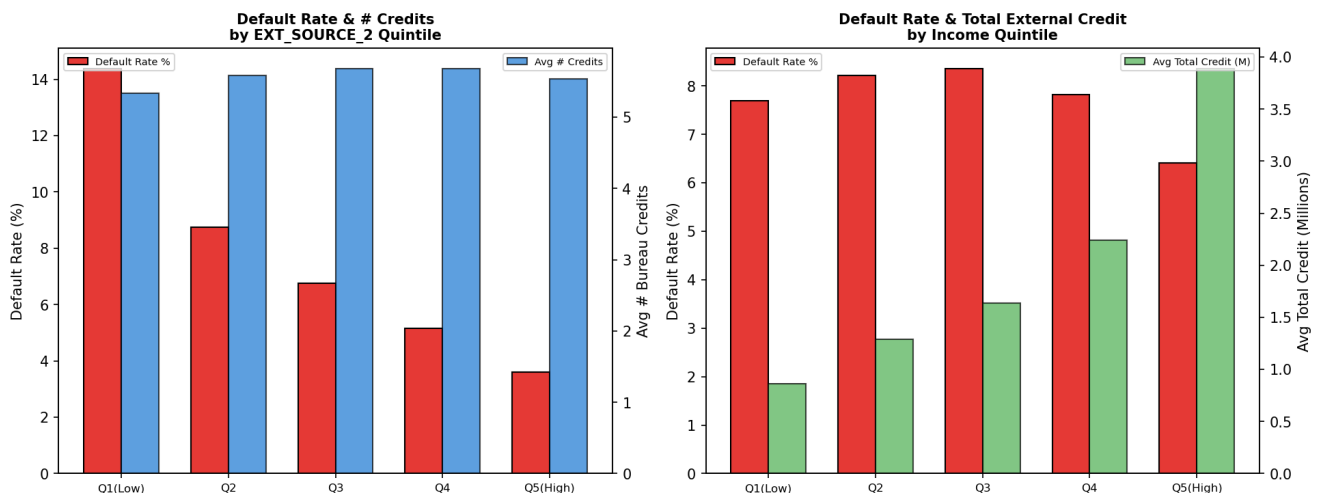
12. Integration with application_train.csv

Bureau.csv has a **direct one-to-many join** to application_train via SK_ID_CURR — the simplest join architecture among supplementary datasets. Each applicant has 0 to 116 bureau credits.

12.1 Coverage & Default Impact

Metric	Value	Implication
Train apps WITH bureau data	263,491 (85.7%)	Strong coverage — most applicants have external history
Train apps WITHOUT bureau data	44,020 (14.3%)	These applicants may lack formal credit history
Default rate WITH bureau	7.73%	Lower — credit experience is protective
Default rate WITHOUT bureau	10.12%	Higher — no credit history is a risk factor
Credits per applicant (mean)	5.6	Multiple credits per person requires aggregation
Credits per applicant (max)	116	Extreme cases need clipping/capping

12.2 Demographic Interaction Analysis



EXT_SOURCE_2 × Bureau credits: Higher external scores correlate with both lower default AND more bureau credits. Well-scored applicants tend to have richer credit histories. This confirms external scores partially capture bureau-derived information — but bureau adds **granularity** (credit type composition, overdue history, debt ratios) that scalar scores cannot represent.

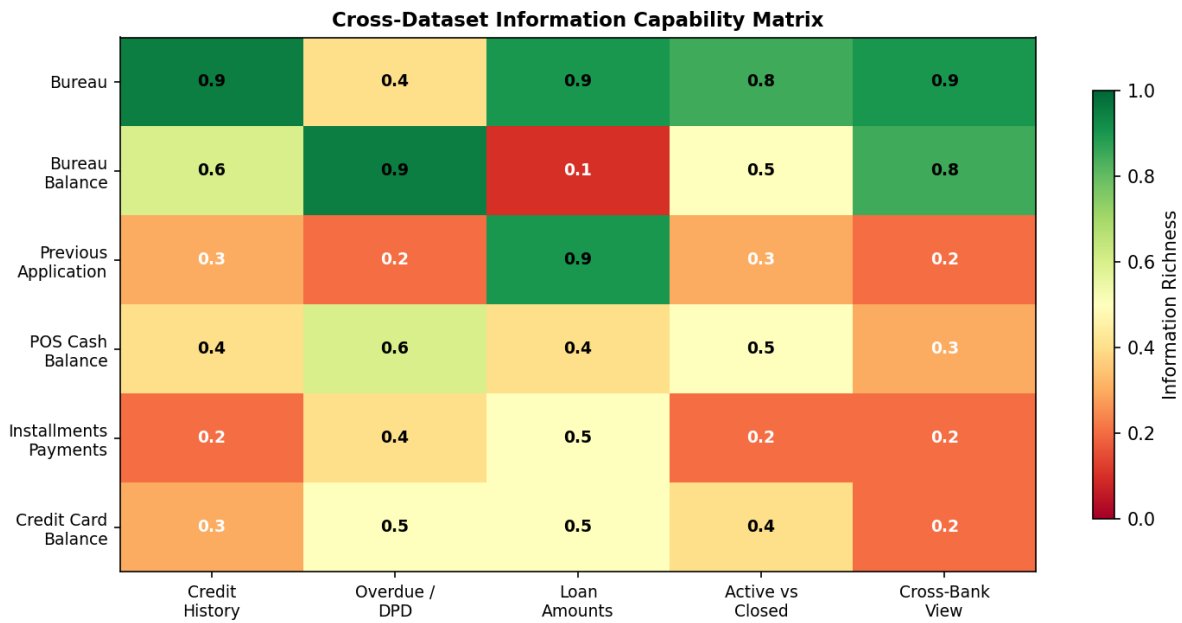
Income × Total external credit: Higher income applicants carry more total external credit — expected given their capacity. However, the default rate relationship with income exists **independently** of credit volume, suggesting income and bureau features provide complementary risk signals.

12.3 Aggregation Strategy

Approach	Logic	Example Features
Simple aggregation	agg(sum/mean/max/min/count) per SK_ID_CURR	n_credits, total_debt, max_overdue, avg_credit_sum

Approach	Logic	Example Features
Status segmentation	Compute features separately for Active vs Closed credits	n_active_credits, total_active_debt, active_credit_mean_amount
Type segmentation	Features per credit type (consumer, card, etc.)	n_consumer_credits, has_microloan, pct_revolving_credit
Temporal segmentation	Split by recency: last 1yr / 1-3yr / 3yr+	n_recent_credits, recent_overdue, old_vs_new_credit_ratio
Ratio features	Cross-column ratios and comparisons	debt_to_credit_ratio, active_to_closed_ratio, overdue_to_credit_pct

13. Cross-Dataset Synergies



13.1 Bureau + Bureau Balance (Parent-Child)

Bureau.csv provides **credit-level metadata** (type, amount, status, timeline). Bureau Balance adds **monthly behavioral time-series**. Together: *'This 500K consumer credit, opened 2 years ago and currently active, has shown DPD-1 in 3 of the last 6 months and is escalating.'* Neither dataset alone provides this complete picture.

13.2 Bureau + Previous Application

Previous Application shows Home Credit's own decisions; Bureau shows external credit activity. **Strategic comparison:** Was an applicant rejected by Home Credit while simultaneously taking 3 new consumer credits elsewhere? This reveals **credit-seeking behavior** — rejected applicants who immediately open external credits may be desperate for funding.

13.3 Bureau + Internal Behavioral Data

Bureau's AMT_CREDIT_SUM and AMT_CREDIT_SUM_DEBT enable **total cross-institutional debt calculation**. Combined with Home Credit's own AMT_CREDIT from application_train and POS/CC balance data, you can compute the applicant's **total financial exposure** across ALL institutions — the gold standard for debt-to-income analysis.

Dataset	How Bureau Helps It	How It Helps Bureau
application_train	External credit portfolio: amounts, types, overdue, timeline, leverage	TARGET label, demographics, EXT_SOURCE for validation
bureau_balance	Credit metadata: type, amount, status to contextualize monthly behavior	Monthly behavioral dynamics for temporal feature engineering
previous_application	Was applicant also seeking credit externally? Cross-selling signals	Home Credit decisions provide contrast with external activity
POS Cash / Installments	Total debt calculation: external + internal	Internal payment behavior to compare with external record

Dataset	How Bureau Helps It	How It Helps Bureau
credit_card_balance	External revolving credit context (CC from bureau)	Internal revolving behavior for cross-product comparison

Bureau.csv's unique position: It is the ONLY dataset providing (1) external credit amounts, (2) external credit types, (3) external active/closed status, (4) external overdue history amounts. All internal datasets (POS, Installments, CC) only show Home Credit's own portfolio. Bureau is the **external lens** that completes the 360-degree view of the applicant.

14. Feature Engineering Playbook

#	Feature Name	Logic	Expected Value
1	bur_n_credits	Count distinct SK_ID_BUREAU per applicant	Medium — credit experience count
2	bur_n_active	Count credits with CREDIT_ACTIVE = Active	High — current external obligation count
3	bur_active_ratio	n_active / n_credits	Very High — strongest bureau signal; monotonic with default
4	bur_has_bad_debt	Binary: any CREDIT_ACTIVE = Bad debt	Very High — rare but devastating; 20% default rate
5	bur_total_debt	Sum of AMT_CREDIT_SUM_DEBT for active credits	High — total external exposure
6	bur_debt_to_credit	total_debt / total_credit	High — external leverage ratio
7	bur_max_overdue_amt	Max of AMT_CREDIT_MAX_OVERDUE	High — worst historical overdue
8	bur_has_current_overdue	Binary: any CREDIT_DAY_OVERDUE > 0	Very High — active external delinquency
9	bur_days_since_first	-min(DAYS_CREDIT): credit history length	Medium — experience duration
10	bur_days_since_last	-max(DAYS_CREDIT): most recent credit	Medium — recency of credit activity
11	bur_has_microloan	Binary: any CREDIT_TYPE = Microloan	High — distress signal; 20.6% default rate
12	bur_pct_consumer	Proportion of consumer credit vs total	Medium — credit composition
13	bur_n_prolonged	Sum of CNT_CREDIT_PROLONG	Medium — inability to complete terms on schedule
14	bur_avg_credit_update	Mean DAYS_CREDIT_UPDATE	Medium — recency of bureau data
15	bur_total_annuity	Sum of AMT_ANNUITY (active credits)	High — monthly obligation load at external institutions
16	bur_early_close_rate	% of closed credits where DAYS_ENDDATE_FACT < DAYS_CREDIT_ENDDATE	Medium — prepayment tendency; positive behavioral signal
17	bur_n_credits_1yr	Credits opened in last year (DAYS_CREDIT > -365)	High — recent credit-seeking behavior; velocity
18	bur_pct_revolving	Proportion of credit card / revolving types	Medium — revolving dependency
19	bur_overdue_to_debt	AMT_CREDIT_SUM_OVERDUE / AMT_CREDIT_SUM_DEBT	High — what fraction of debt is past due
20	bur_max_credit_amount	Max AMT_CREDIT_SUM	Low-Med — scale of largest external credit

Advanced Feature Strategies:

- * **Temporal decay weighting:** Weight features by recency. A credit opened 6 months ago matters more than one from 7 years ago. Use exponential decay: $\text{weight} = \exp(\text{DAYS_CREDIT} / 365)$. This makes recent activity more influential in aggregated features.
- * **Active-only vs All-credit features:** Compute features separately for active credits (current obligations) and all credits (historical pattern). The ratio between active-feature and all-feature captures changes in behavior.
- * **Credit velocity:** Count credits opened per year. Increasing velocity (more credits recently) may signal increasing financial need. Decreasing velocity may signal credit access tightening.
- * **Cross-institution debt load:** Sum active `AMT_CREDIT_SUM_DEBT` from bureau + `AMT_CREDIT` from `application_train` = total debt exposure. Compare to `AMT_INCOME_TOTAL` for comprehensive DTI.

15. Business Recommendations

15.1 For Credit Risk Team

External leverage screening: Reject or downgrade applicants with (1) `active_ratio` > 0.8, (2) external debt-to-credit > 0.9, (3) any current `CREDIT_DAY_OVERDUE` > 0, or (4) any Bad debt status. These four conditions capture the highest-risk segments.

- * **Microloan alert system:** Flag all applicants with external microloan history for enhanced due diligence. With 20.6% default rate, microloans are the strongest single credit-type risk signal. Consider requiring additional documentation or guarantors for these applicants.
- * **Credit velocity monitoring:** Track the rate of new external credit applications (credits opened in last 6/12 months). Accelerating credit-seeking behavior often precedes financial distress by 3-6 months.

15.2 For Product Team

- * **Premium segment identification:** Applicants with all-closed external portfolios, zero overdue history, and 5+ years of credit history are proven creditworthy (5.7% default vs 8.1% baseline). Offer these customers preferred interest rates and larger credit limits to acquire profitable, low-risk business.
- * **Cross-sell opportunities:** Applicants with external credit cards but no Home Credit revolving product represent cross-selling targets. Their credit card behavior at other institutions (via bureau) validates their revolving credit management ability.

15.3 For Data Science Team

- * **Top-5 features:** `bur_active_ratio`, `bur_has_bad_debt`, `bur_has_current_overdue`, `bur_debt_to_credit`, `bur_has_microloan`. These are the highest-impact bureau features based on default rate separation and correlation analysis.
- * **Feature interaction priorities:** (1) `active_ratio` × `EXT_SOURCE_2`; (2) `total_debt` × `AMT_INCOME_TOTAL` (= external debt-to-income); (3) `n_credits_1yr` × `age`. Bureau features interact powerfully with demographic and scoring variables.
- * **Handle the 14.3% without bureau data:** (1) `has_bureau_data` flag; (2) fill NULLs with -999 for tree models; (3) explore target-encoded substitution using demographic proxies. The no-bureau segment has distinct characteristics worth separate modeling.

16. Conclusions

Dimension	Finding	Action
Dataset Role	Most information-rich supplementary dataset: 17 columns, 1,716,428 credits	Central to external risk assessment; highest feature yield
Coverage	85.7% of train apps; 85.7% have bureau history	Strong coverage; flag the 14.3% without bureau data
Top Signal	Active/closed ratio: All-closed = 5.7% default; All-active = 11.4%	bur_active_ratio is the #1 bureau predictive feature
Risk Flags	Microloan = 20.6% default; Bad debt = 20.0% default	has_microloan, has_bad_debt: rare but powerful binary flags
Financial Metrics	Credit amounts, debt, overdue, annuity — full exposure view	debt_to_credit ratio, total_active_debt, max_overdue
Temporal Depth	Up to 8 years of credit history; median ~2.7 years	Recency-weighted features; credit velocity analysis
Data Quality	7 columns with 6-71% nulls; extreme outliers in amounts	Null-as-feature flags; winsorization; median aggregations
Cross-Dataset	Bridge to bureau_balance; provides amounts no other dataset has	External+internal debt load; total exposure calculation
Feature Yield	20+ engineered features across 5 strategies	Expect +0.01-0.02 AUC from bureau features alone

Final Assessment: Bureau.csv is the **crown jewel of supplementary datasets**. With 17 columns spanning credit amounts, types, overdue history, and lifecycle status — all from external institutions — it provides a **360-degree external credit view** unavailable from any internal dataset. Its 85.7% train coverage is the second highest among supplementary tables. The **active/closed ratio** alone separates 5.7% from 11.4% default rates — a 2× risk differential. Credit type profiling (microloans at 20.6% default) and the debt-to-credit ratio provide additional orthogonal risk dimensions. Bureau features typically contribute **+0.01-0.02 AUC** improvement and consistently rank in the top 20 features of champion LightGBM models. Combined with its child dataset (bureau_balance), the bureau data family provides the most powerful external risk signal in the Home Credit feature ecosystem.